

AI USAGE

Overall AI Architecture

User Request



Orchestrator Agent



Knowledge Retrieval Agent



Dependency Agent



Script Generation Agent



Security Review Agent



Validation Agent



Troubleshooting Agent



Documentation Agent



Incident Agent



Analytics Agent



Final Output

1. Orchestrator Agent (AI Brain)

AI Usage

This agent uses an LLM to understand user intent and determine:

What software user wants

Which OS is being used

Which agents should execute

Execution order

Example

User Input:

Install Docker on Ubuntu 22.04

AI extracts:

```
{  
  "software": "Docker",  
  "os": "Ubuntu",  
  "version": "22.04",  
  "package_manager": "apt"  
}
```

Why AI?

Without AI:

Hardcoded If Else

With AI:

Natural Language Understanding

User can type:

Need Docker on Ubuntu

or

Can you setup docker?

AI understands both.

2. Knowledge Retrieval Agent (RAG AI)

This is one of the strongest AI features.

Problem

Installation instructions constantly change.

Hardcoding commands is dangerous.

AI Solution

Use:

Gemini

+

ChromaDB

+

Embeddings

Workflow

Official Docs

PDFs

Blogs

Knowledge Base



Embeddings



ChromaDB



Semantic Search

User asks:

Install Docker Ubuntu

AI searches semantically.

Not keyword matching.

Example

AI understands:

Docker installation

and

Container runtime setup

may mean similar things.

3. Dependency Detection Agent

AI Usage

Uses LLM reasoning.

User:

Install Kubernetes

AI thinks:

Kubernetes needs:

Docker

Containerd

Kubectrl

Kubeadm

Kubelet

Generated Report

Dependencies Found

✓ Containerd

✓ Kubeadm

✓ Kubectrl

✓ Kubelet

Traditional systems need manually coded dependency trees.

AI dynamically reasons.

4. Script Generation Agent

This is the project's core AI feature.

Input

Software: Docker

OS: Ubuntu 22.04

Package Manager: apt

AI generates

`sudo apt update`

`sudo apt install docker.io -y`

`sudo systemctl enable docker`

`sudo systemctl start docker`

Supported Outputs

Bash

PowerShell

Batch

Docker Commands

Kubernetes YAML

AI understands:

Ubuntu ≠ Windows

and generates different scripts.

5. Security Review Agent

One of the most unique AI features.

AI scans generated scripts.

Looks for:

`rm -rf`

`chmod 777`

`curl unknown URLs`

`wget unknown domains`

Privilege escalation

Unsafe permissions

Example

Script:

`chmod 777 /`

AI detects:

Critical Security Risk

Output

Risk Score: 9/10

AI acts like a DevSecOps Engineer.

6. Validation Agent

AI Usage

Validates generated scripts.

Checks:

Syntax

OS compatibility

Package manager

Dependency availability

Example

Ubuntu + apt

Valid

Example

Windows + apt

Invalid

AI identifies mismatch.

7. Troubleshooting Agent

This makes the project extremely practical.

User Error:

```text

docker command not found

AI analyses logs.

Determines:

Docker service not started

Provides:

sudo systemctl start docker

AI behaves like a support engineer.

### 8. Documentation Agent

Uses AI to automatically create documentation.

Generates:

Installation Guide

Verification Guide

Rollback Guide

FAQs

Troubleshooting Steps

Example

Input:

Docker Installation

Output:

Step 1

Install package

Step 2

Start service

Verification

docker --version

Rollback

sudo apt remove docker

No manual documentation writing.

## **9. Incident Ticket Agent**

**AI automatically classifies issues.**

**Input**

**Docker Installation Failed**

**AI predicts:**

**Priority: High**

**Category:**

**Infrastructure**

**Severity:**

**Critical**

**Creates:**

**INC-2026-001**

**AI performs ITSM automation.**

## **10. Analytics Agent**

**AI converts raw data into insights.**

**Tracks:**

**Most Installed Software**

**Success Rate**

**Failure Rate**

**Popular OS**

**Popular Package Manager**

**AI generates insights:**

**Docker accounts for 45% of requests.**

**Ubuntu has highest success rate.**

**Windows failures increased by 12%.**

**This becomes an AI-powered reporting system.**

## Agent Memory (Advanced AI Feature)

The system remembers previous executions.

Example

Previous Request:

Docker Ubuntu

Later:

Install Kubernetes

AI remembers:

Docker already installed

and skips unnecessary steps.

Conversational AI Assistant

Built using Gemini/Ollama.

User:

## Why did installation fail?

AI:

The Docker daemon is not running.

Start the service using:

```
sudo systemctl start docker
```

AI Technologies Used

Component AI Technology

Intent Understanding Gemini

Agent Reasoning Gemini + LangGraph

Knowledge Search RAG

Embeddings Ollama Embeddings

Semantic Search ChromaDB

Script Generation Gemini

Security Analysis Gemini

Troubleshooting Gemini

Documentation Generation Gemini

Analytics Insights Gemini